EXPERIMENT NO 6

AIM: To Analyse the Function of Transistor as an Amplifier Using PSpice

PROBLEM STATEMENT: To analyse the function of a transistor as an amplifier using PSpice and determine the voltage gain of the common-emitter (CE) amplifier.

SOFTWARE REQUIRED: Pspice Simulator

COMPONENTS REQUIRED:

NPN Transistor	2N2222 / BC547*
Collector resistor	4.7 kΩ
Emitter resistor	220 Ω
Load resistor	47 kΩ
Resistor (R1)	68 kΩ
Resistor (R2)	2.2 k Ω
Coupling capacitor (C1)	2.2 µF
Coupling capacitor (C_L)	2.2 µF
Emitter bypass capacitor	100 µF
DC supply	24 V
Input source	Sine wave

THEORY:

A transistor can be used as an amplifier to increase the amplitude of a weak electrical signal. In an amplifier circuit, a small AC signal is applied at the input of the transistor, and a larger AC signal is obtained at the output. The transistor does not create energy; instead, it uses the DC supply energy to provide an amplified output signal.

A Common Emitter (CE) configuration is widely used for transistor amplification because it provides both good voltage gain and current gain.

For a transistor operating in the active region:

$$I_C = \beta I_B$$

where
I_C= collector current
I_B= base current
β= current gain of the transistor.

Voltage Gain

The voltage gain of an amplifier is defined as:

$$A_v = \frac{V_o}{V_i}$$

where:

Av= voltage gain

Vo= output voltage

Vi= input voltage

For a simple CE amplifier, the approximate voltage gain is:

$$A_v \approx -g_m R_C$$

Where g_m is the transconductance of the transistor.

The negative sign indicates that the output voltage is 180° out of phase with the input voltage.

CIRCUIT DIAGRAM:

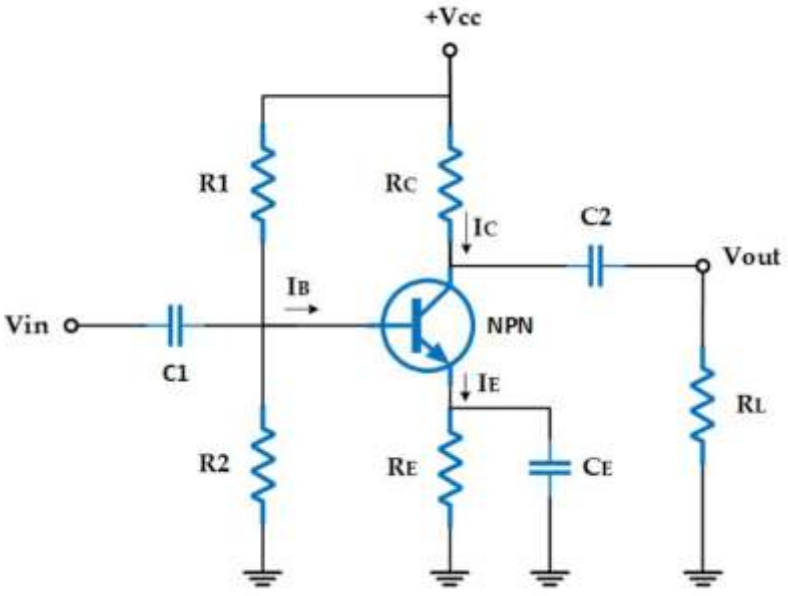


Fig: Common Emitter Amplifier

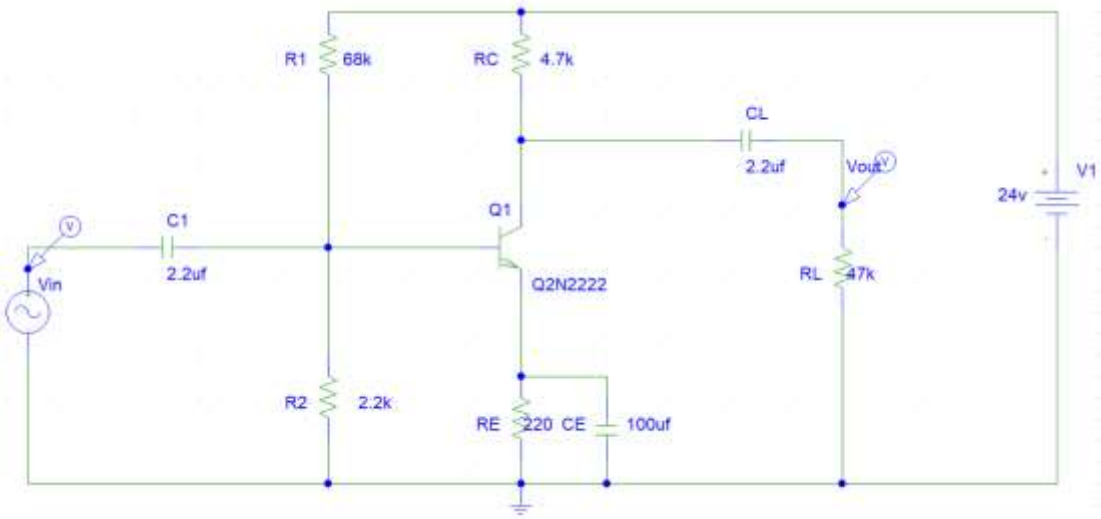


Fig: Common Emitter Amplifier using Pspice

PROCEDURE:

1.Create a new project Select File → New → Project.

2.Place components and make connection as per circuit diagram

3.Set the input source

VSIN source:

- VOFF = 0 V
- VAMPL = 10mV
- FREQ = 1kHz

4. Set up Transient Analysis

- Print step: 0 ms
- Final Time: 5ms
- Step Ceiling: 1us

5.Run the simulation.

OBSERVATION:

Input Voltage (Vin)	
Output Voltage (Vout)	
Input Frequency	
Voltage Gain (Av)	
Phase Difference	

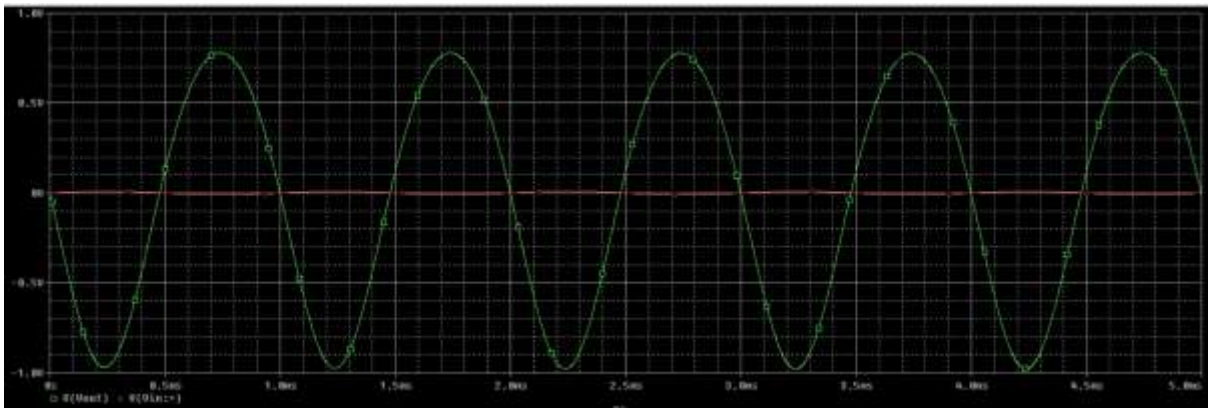
CALCULATIONS:

Voltage gain is:

$$A_v = \frac{V_{out}}{V_{in}}$$

Vin= ___ V ; Vout= ___ V ; Calculate Av.

OUTPUT:



RESULT:

The transistor successfully operates as a voltage amplifier in the common-emitter configuration. The output signal is amplified and is approximately 180° out of phase with the input signal. The voltage gain is calculated from the ratio of output voltage to input voltage.

VIVA VOCE:

1. Why is a transistor used as an amplifier?
2. What is the function of biasing in a transistor amplifier?
3. What is the Q-point?
4. What happens if the transistor enters saturation or cutoff during amplification?
5. Why is the output of a CE amplifier phase shifted by 180°?